



PROJECT DEMONSTRATION



Scalability & Future Plan

 DATE
17 7/9/2026

 NAME
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 PROJECT
OptiCrop – Smart
Agricultural
Production
Optimization Engine

 MAX MARKS
1 Marks

Overview

This document outlines how the current project solution can be scaled to larger user bases, increased data loads, or extended features in the future.

Current System Limitations

LIMITATION	IMPACT	PRIORITY TO ADDRESS
No user authentication	Unauthorized access may occur	High
No database integration	Recommendation history is not stored	Medium
Limited deployment (local Flask app)	Restricted scalability and accessibility	High

Scalability Plan

SCALABILITY ASPECT	CURRENT STATE	PROPOSED UPGRADE / SOLUTION
User Load	Supports limited local users	Deploy on cloud with load balancing
Data Storage	No persistent database	Integrate MySQL/PostgreSQL
Performance	Single-model local execution	Optimize model and enable caching
Security	Basic input validation	Add authentication, HTTPS, and role-based access



Future Roadmap

PHASE	PLANNED FEATURE / ENHANCEMENT	TARGET TIMELINE	EXPECTED IMPACT
Phase 1	Cloud deployment and database integration	1–2 Months	Improved accessibility and data management
Phase 2	User authentication and admin dashboard	2–3 Months	Enhanced security and administration
Phase 3	Recommendation history and analytics dashboard	3–4 Months	Better monitoring and reporting
Phase 4	Advanced ML models (Random Forest, XGBoost) and API integration	4–6 Months	Higher accuracy and enterprise scalability